“Arrow pushing” diagrams are used to describe reaction mechanism

Arrow pushing diagrams follow the following rules:

1. Each arrow represents the movement of one *pair* of valence electrons.
2. Each arrow begins where the electrons originate (either a lone pair of electrons or a bond) and ends where the electrons are going (either an atom or a bond).
3. An arrow that *starts* at an atom represents the movement of a lone pair of electrons.
4. An arrow that *starts* at the center of a bond represents the breaking of that bond.
5. An arrow that *ends* at an atom represents the formation of a new covalent bond or a new lone pair.
6. An arrow that *ends* at a bond represents adding of a bond to form a double (or triple) bond.

In the upcoming lecture we will explore the structural and chemical properties of RNA. In particular, we will examine reactions that are catalyzed by RNA molecules. Before we can discuss these reactions we need to introduce a formalism for describing the movements of electrons that take place during chemical reactions. As we have seen already, a chemical reaction is a rearrangement of valence electrons that leads to the cleavage and/or formation of bonds. The specific electron movements associated with each reaction are described by the **reaction mechanism**. Reaction mechanisms show where electrons originate in the reactant and where they ultimately reside in the product. This is represented graphically using a formalized system called **arrow pushing** in which the movement of electrons is diagrammed using a series of curved arrows. Arrow pushing diagrams adhere to the following rules:

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2. Each arrow begins where the pair of electrons originates (either a lone pair of electrons or a bond) and ends where the electrons are going (either an atom or a bond).
3. An arrow that *starts* at an atom represents the movement of a lone pair of electrons.
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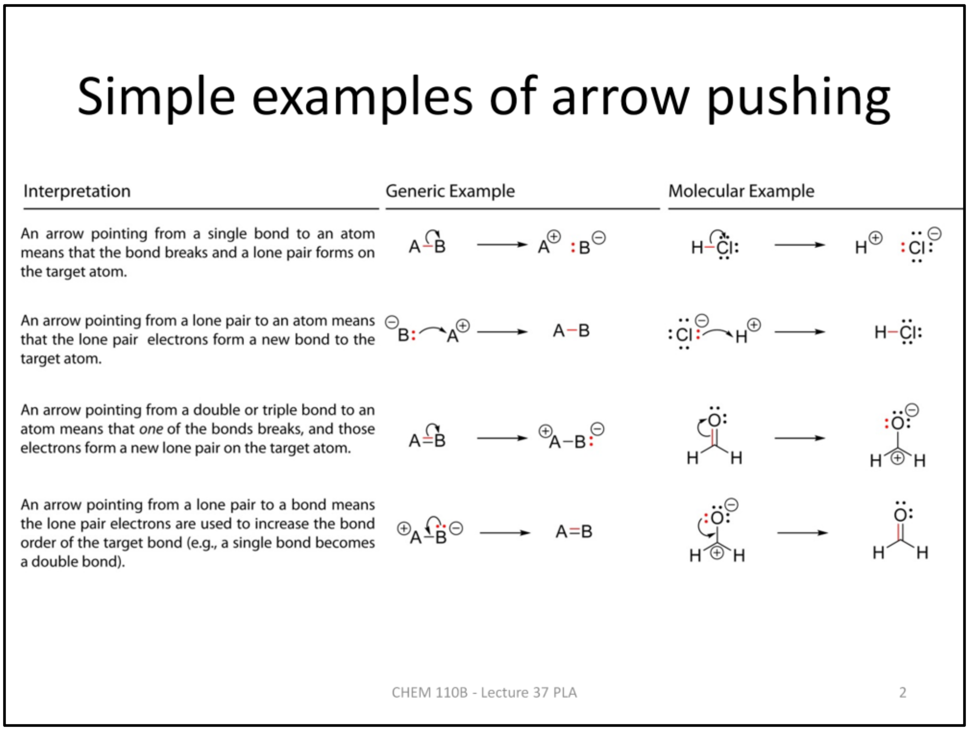
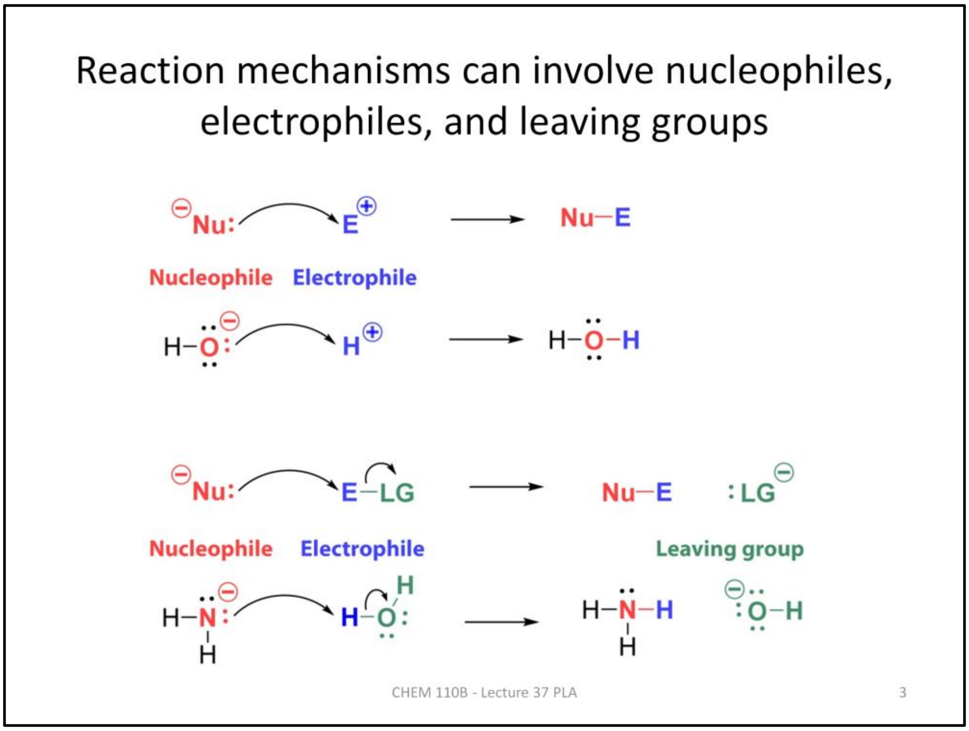


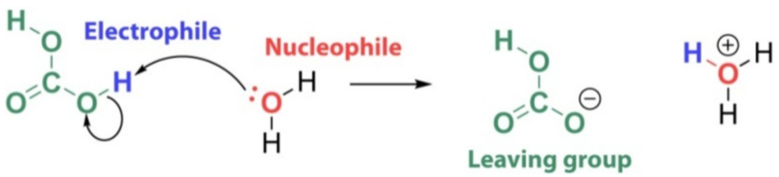
Applying the rules of arrow pushing so that they become second nature requires practice. This slide shows examples of mechanisms that are described by individual arrows. A few things worth pointing out are that the lone pair/bond moving is colored red in each example, and there is a separation of charge in either the reactants or products in each example.



As we have seen, many molecules contain atoms that carry full or partial charges. Reactions between molecules (small molecules or functional groups on macromolecules) often occur when the electrons from an atom carrying a negative charge (full or partial) forms a bond with an atom in another molecule that carries a positive charge (full or partial). For reactions in which a bond is formed between atoms that were not previously bonded, the electron-rich atom from which the electrons originate is known as the **nucleophile** (Nu:) and the electron-deficient atom is known as the **electrophile** (E) (shown on the slide). As their names imply, nucleophiles are attracted to nuclei and electrophiles are attracted to electrons (*philos* being Greek for love).

Another concept that is important in understanding reaction mechanisms is the **leaving group** (LG). This term is used to describe an atom or portion of a molecule that separates from a molecule as a result of a bond breaking during the course of a reaction. Leaving groups always take both electrons from the broken bond with them as they depart. Note that there are two sets of reactions here. The first set of reactions is a combination reaction, the second set of reactions is a single displacement reaction and so has a leaving group, which is colored green. Also note that charge is the same on the left (reactant state) and right (product state) of the reaction.

Example: Arrow pushing diagram for the dissociation of carbonic acid



Let us apply these concepts to the reaction of carbonic acid (H_2CO_3) with water to yield bicarbonate and a hydronium ion. The reaction mechanism is described using two arrows (see figure on slide). One arrow points from the lone pair of electrons on the water molecule to a hydrogen atom on carbonic acid. This arrow conveys the idea that the oxygen from water is using its lone pair electrons to form a bond with the hydrogen atom. Since hydrogen can only form one bond at a time, it must break its existing bond to oxygen. Breaking the existing O-H bond is represented by the second arrow, which points from the O-H bond in carbonic acid to the oxygen atom in the same O-H bond. This arrow conveys the idea that the O-H bond is breaking and that the electrons from that bond are being retained by carbonic acid's oxygen as a lone pair. In this reaction the oxygen in water is a nucleophile and the hydrogen in carbonic acid is an electrophile, as the oxygen in water is using its electrons to form a new bond and hydrogen is the recipient of those electrons. Finally, the remainder of the carbonic acid molecule (HCO_3^-) serves as a leaving group, as it results from the cleavage of the carbonic acid O-H bond and it takes a pair of electrons with it. Note that charge is balanced on both sides of the reaction.